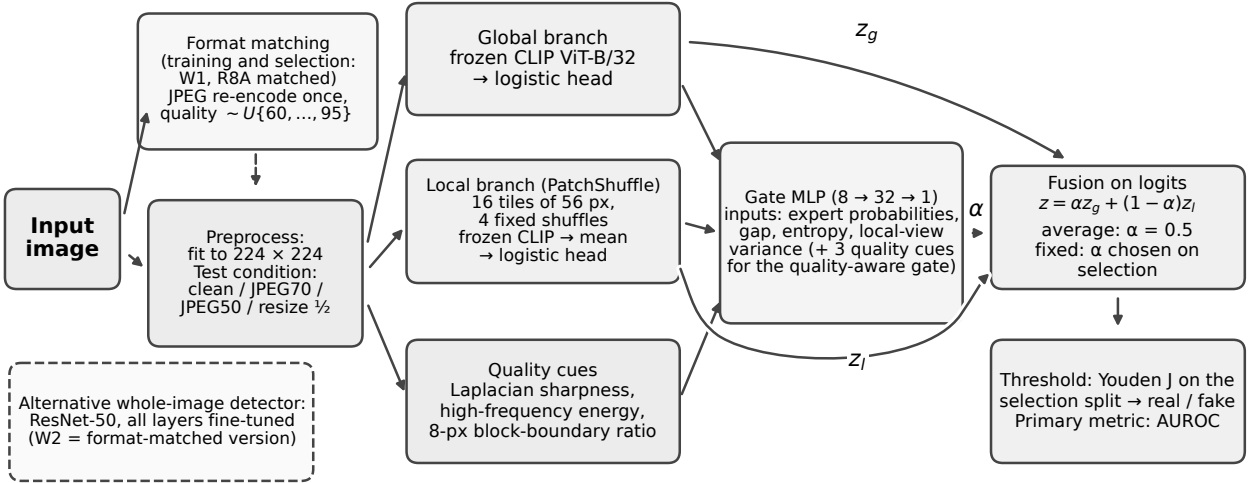
Input image
Format matching (training and selection: W1, R8A matched) JPEG re-encode once, quality $\sim U\{60, \ldots, 95\}$
Preprocess: fit to 224 × 224 Test condition: clean / JPEG70 / JPEG50 / resize ½
Global branch frozen CLIP ViT-B/32 → logistic head
Local branch (PatchShuffle) 16 tiles of 56 px, 4 fixed shuffles frozen CLIP → mean → logistic head
Quality cues Laplacian sharpness, high-frequency energy, 8-px block-boundary ratio
z_g
Gate MLP (8 → 32 → 1) inputs: expert probabilities, gap, entropy, local-view variance (+ 3 quality cues for the quality-aware gate)
α
z_l
Fusion on logits $z = \alpha z_g + (1-\alpha) z_l$ average: α = 0.5 fixed: α chosen on selection
Threshold: Youden J on the selection split → real / fake Primary metric: AUROC
Alternative whole-image detector: ResNet-50, all layers fine-tuned (W2 = format-matched version)